

Analytic Proof of the Prime Number Theorem via Newman's Tauberian Theorem

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Abstract

This document outlines the specific analytic proof of the Prime Number Theorem (PNT) that we intend to formalize in Lean 4. It is based on D.J. Newman's elegant complex integration method, popularized by Don Zagier, which completely avoids the heavy machinery of the Wiener-Ikehara Tauberian theorem.

1 Introduction

The Prime Number Theorem states that $\pi(x) \sim \frac{x}{\log x}$, or equivalently, that Chebyshev's theta function $\vartheta(x) = \sum_{p \leq x} \log p$ satisfies $\vartheta(x) \sim x$.

2 Newman's Tauberian Theorem

Theorem 1 (Newman 1980). *Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a bounded, locally integrable function. Its Laplace transform*

$$g(z) = \int_0^\infty f(t)e^{-zt} dt$$

is analytic on $\Re(z) > 0$. If $g(z)$ can be analytically continued to the closed right half-plane $\Re(z) \geq 0$, then the improper integral $\int_0^\infty f(t) dt$ converges and is equal to $g(0)$.

Proof. For $T > 0$, define the truncated integral $g_T(z) = \int_0^T f(t)e^{-zt} dt$. This $g_T(z)$ is an entire function. We must show $\lim_{T \rightarrow \infty} g_T(0) = g(0)$.

By Cauchy's Integral Formula, for a path C , we can write:

$$g(0) - g_T(0) = \frac{1}{2\pi i} \int_C (g(z) - g_T(z)) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z}$$

Let $R > 0$. Since g is analytic on $\Re(z) \geq 0$, it is analytic on a slightly larger region $\Re(z) \geq -\delta$ for some $\delta > 0$ (depending on R). We choose the contour C to be the boundary of the region $\{z \in \mathbb{C} : |z| \leq R, \Re(z) \geq -\delta\}$.

The boundary C consists of:

- C_+ : the right semicircle arc $|z| = R, \Re(z) > 0$.
- C_- : the left semicircle arc inside the analytic region $|z| = R, \Re(z) \leq 0, \Re(z) \geq -\delta$.
- Vertical line segments connecting the arcs.

By bounding the integrand on the semicircle C_+ and C_- separately:

1. **On C_+ :** Here $|z| = R$ and $\Re(z) > 0$. The geometric factor simplifies cleanly:

$$\left(\frac{1}{z} + \frac{z}{R^2}\right) = \frac{\bar{z} + z}{R^2} = \frac{2\Re(z)}{R^2}$$

This directly **cancels the growth** of the exponential $e^{zT} = e^{\Re(z)T}$. Using the bound $|f(t)| \leq M$, we find the integral over C_+ is strictly bounded by $O(1/R)$.

2. **On C_- :** We split the integrand. Since $g_T(z)$ is entire, its integral can be shifted to the full left semicircle where e^{zT} decays exponentially rapidly as $T \rightarrow \infty$. For $g(z)$, which is independent of T , its integral over C_- goes to zero as $T \rightarrow \infty$ by basic dominated convergence.

Thus, taking $\limsup_{T \rightarrow \infty}$ and then making R arbitrarily large, the absolute difference $|g(0) - g_T(0)|$ vanishes. $\square$

3 The Riemann Zeta Function and Properties

We recall the Riemann Zeta function for $\Re(s) > 1$:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}$$

Taking the logarithmic derivative:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{\log p}{p^s - 1}$$

By standard methods (subtracting the simple pole at $s = 1$), the function $\zeta(s) - \frac{1}{s-1}$ extends analytically to $\Re(s) > 0$. Crucially, using the identity $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$, one proves that $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$.

4 Proof of the Prime Number Theorem

Lemma 1. *Chebyshev's bound holds: $\vartheta(x) \leq Cx$ for some constant $C > 0$.*

Define the target function $f(t) = \vartheta(e^t)e^{-t} - 1$. Since $\vartheta(x) = O(x)$, $f(t)$ is strictly bounded. The Laplace transform of $f(t)$ evaluates to:

$$g(z) = \int_0^\infty (\vartheta(e^t)e^{-t} - 1)e^{-zt} dt = \frac{1}{z+1}\Phi(z+1) - \frac{1}{z}$$

where $\Phi(s) = \sum_p \frac{\log p}{p^s}$.

Since $\Phi(s)$ differs from $-\frac{\zeta'(s)}{\zeta(s)}$ by a function analytic for $\Re(s) > 1/2$, and $-\frac{\zeta'(s)}{\zeta(s)}$ has a simple pole at $s = 1$ with residue 1, the pole of $\Phi(z+1)$ at $z = 0$ exactly cancels the $-1/z$ term. Furthermore, since $\zeta(1+it) \neq 0$, $\Phi(z+1)$ has no other poles on the imaginary axis $\Re(z) = 0$.

Therefore, $g(z)$ extends analytically to the closed right half-plane $\Re(z) \geq 0$. By Newman's Tauberian Theorem, the improper integral

$$\int_0^\infty (\vartheta(e^t)e^{-t} - 1) dt$$

must converge, which forces $\lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1$. This completes the proof of the Prime Number Theorem.

Expansion Layer: The Contour Modifier

You clicked on the Tauberian contour modification: $\left(1 + \frac{z^2}{R^2}\right)$.

In the classical proof, this factor elegantly binds the integral on the semicircle boundary. However, in our formalization, we must avoid invoking heavy Lebesgue measure theory purely to bind integral limits.

Instead of dealing with integrals immediately, we [isolate Newman's modifier into a standalone algebraic definition](#). By bounding the algebraic identity strictly within $\mathbb{C}$, we prove absolute limits on the circular boundary analytically before the integral is ever evaluated.

→ [\[Click here to view the Line-by-Line Lean Implementation\]](#) ←

Expansion Layer: Finite Zeta Limits

You clicked on the infinite Euler product: $\prod_p \left(1 - \frac{1}{p^s}\right)^{-1}$.

Infinite products $\prod_p$ over all primes require complex topological definitions of convergence that are extremely brittle in proof assistants. To keep the proof constructive and contained, we replace the infinite topological limit with a [classical epsilon-delta bounded Finset projection](#).

We bound the expansion into a strict N -truncation (the set of all primes less than or equal to N), evaluating the geometric expansion precisely over finite sums.

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Expansion Layer: Bridging the Fundamental Theorem of Arithmetic

You clicked on the infinite integer sum: $\sum_{n=1}^{\infty} \frac{1}{n^s}$.

The paper implicitly claims that expanding the prime Euler product generates the sum over all integers $\mathbb{N}$. In Lean, we cannot "implicitly" expand an infinite product.

We must formally bridge the combinatorial space of prime power sequences (**Finsupp**) to standard integers. To do this without invoking measure theory, we define a [bounded smooth numbers subset](#) $S(N, K)$ representing primes up to N with powers up to K . We then prove a strict bijection showing that the geometric expansion maps perfectly into this finite subset.

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Expansion Layer: The Finite Chebyshev Bound

You clicked on the Chebyshev target function: $\vartheta(e^t)e^{-t} - 1$.

Newman leverages Chebyshev's function $\vartheta(x) = \sum_{p \leq x} \ln p$. In analysis, evaluating integrals of step-functions bounded by limits requires dense real analysis.

For our formalization, we [define \$\vartheta\(x\)\$ explicitly using `Finset.sum`](#) over filtered primes. This locks the limit bounds directly into array-sum equivalencies rather than topological measures.

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Code Layer: Algebraic Modifiers in Contour.lean

You are viewing the granular Lean 4 code interspersed with mathematical logic for Newman's Modifier.

First, we define the modifier identically to the paper, operating over the complex field $\mathbb{C}$.

```
noncomputable def newmanModifier (s :  $\mathbb{C}$ ) (R :  $\mathbb{R}$ ) :  $\mathbb{C}$  :=  
  (1 :  $\mathbb{C}$ ) + s^2 / (R^2 :  $\mathbb{C}$ )
```

Next, we establish a 50-line sub-lemma explicitly verifying the magnitude reduction on the boundary where $|s| = R$. We prove this purely algebraically.

```
lemma newmanModifier_eq_on_circle (s :  $\mathbb{C}$ ) (R :  $\mathbb{R}$ ) (hR : R > 0) (hs :  
  Complex.abs s = R) :  
  newmanModifier s R = (s + R^2 / s) / s
```

Code Layer: Bounding the Zeta Product in Zeta.lean

You are viewing the granular Lean 4 code for the bounded Euler Product.

We define the partial product up to a finite integer N .

```
noncomputable def eulerPartialProduct (s : ℂ) (N : ℕ) : ℂ :=  
  ∏ p ∈ (Finset.range N).filter Nat.Prime, (1 - ((p : ℂ) ^ (-s)))-1
```

Then, we prove the finite geometric series expansion for each individual prime factor up to a bounded power K . This avoids evaluating the limit to infinity.

```
lemma prime_euler_factor_inv (p : ℕ) (s : ℂ) (hs : s.re > 0) (K : ℕ) (hp :  
  Nat.Prime p) :  
  (1 - ((p : ℂ) ^ (-s)))-1 = (∑ k ∈ Finset.range K, ((p : ℂ) ^ (-s))k)  
  +  
    (((p : ℂ) ^ (-s))K * (1 - ((p : ℂ) ^ (-s)))-1
```

Code Layer: Combinatorial Bijection in Zeta.lean

You are viewing the granular Lean 4 code mapping the Fundamental Theorem of Arithmetic over bounded sets.

We define N -smooth numbers to isolate integers with prime factors bounded by N .

```
def SmoothNumbers (N : ℕ) : Set ℕ :=
  {n : ℕ | n > 0 ∧ ∀ p, p.Prime → p | n → p ≤ N}
```

We bridge the integer set directly to the prime power representation (Finsupp) via Lean's core arithmetic libraries.

```
lemma smooth_numbers_fta_bijection (N : ℕ) (n : ℕ) (hn : n ∈ SmoothNumbers N) :
  ∃! (f : ℕ →₀ ℕ), (∀ p ∈ f.support, p.Prime ∧ p ≤ N) ∧ f.prod (fun p k
=> p ^ k) = n := by {
  -- The existence is witnessed by the standard prime factorization of n
  use n.factorization
  constructor
  · constructor
    · intro p hp
      have h_prime := Nat.prime_of_mem_primeFactors hp
      have h_dvd := Nat.dvd_of_mem_primeFactors hp
      exact ⟨h_prime, hn.right p h_prime h_dvd⟩
    · exact Nat.prod_factorization_pow_eq_self (ne_of_gt hn.left)
  · -- Uniqueness follows from the equivalence between ℕ+ and prime-
    supported Finsupps
    intro g hg
    have h_prime_supp : ∀ p ∈ g.support, p.Prime := fun p hp => (hg.1 p hp
    ).1
    let g_sub : { f : ℕ →₀ ℕ // ∀ p ∈ f.support, p.Prime } := ⟨g,
    h_prime_supp⟩
    have h_symm : (Nat.factorizationEquiv.symm g_sub).val = n := by {
      have h1 : (Nat.factorizationEquiv.symm g_sub).val = g_sub.val.prod (
      fun p k => p ^ k) :=
        Nat.factorizationEquiv_symm_apply_coe g_sub
      rw [h1]
      exact hg.2
    }
    let n_pos : ℕ+ := ⟨n, hn.left⟩
    have h_symm_eq : Nat.factorizationEquiv.symm g_sub = n_pos := Subtype.
    ext h_symm
    have h_equiv := Equiv.apply_symm_apply Nat.factorizationEquiv g_sub
    rw [h_symm_eq] at h_equiv
    have h_val : (Nat.factorizationEquiv n_pos).val = g_sub.val := by rw [
    h_equiv]
    exact h_val.symm
  }
```

We then declare the strictly bounded mapping allowing us to commute the product logic.

```
noncomputable def smoothNumbersBoundedFinset (N K : ℕ) : Finset ℕ :=
  have : DecidablePred (fun n => n > 0 ∧ ∀ p, p.Prime → p | n → p ≤ N ∧ n.
  factorization p ≤ K) := Classical.decPred _
  (Finset.range ((N.factorial ^ K) + 1)).filter (fun n => n > 0 ∧ ∀ p, p.
  Prime → p | n → p ≤ N ∧ n.factorization p ≤ K)
```

```

lemma euler_product_expand_sum (N K : ℕ) (s : ℂ) :
  ∏ p ∈ (Finset.range (N + 1)).filter Nat.Prime, (∑ k ∈ Finset.range (K
+ 1), ((p : ℂ) ^ (-s)) ^ k) =
  ∑ n ∈ smoothNumbersBoundedFinset N K, (n : ℂ) ^ (-s) := by {
-- This proof requires commuting a product of sums (Finset.prod_sum)
-- and matching each integer with its unique exponent vector via our FTA
  bijection.
  sorry
}

```

Code Layer: Finite Chebyshev Representation

You are viewing the granular Lean 4 code defining the finite Chebyshev sum.

Rather than integrating over prime steps continuously, we filter a strict range and sum the real logarithms directly.

```
noncomputable def chebyshevTheta (x : ℝ) : ℝ :=  
  ∑ p ∈ (Finset.range ⌊x⌋+).filter Nat.Prime, Real.log p
```